

# 6. Project Testing Phase

## ComicCraft – AI Comic Story Creator

### Introduction

The testing phase verifies that the individual ComicCraft components and the complete workflow operate together as required.

### Functional Testing

Test the creation form with valid and invalid story prompts, character information, settings, tones, art styles, and panel counts. Verify that panel count is limited to 1–12.

### API Testing

Verify GET /, GET /api/health, POST /api/generate-story, POST /api/create-comic, POST /api/regenerate-panel, and GET /api/comic/{comic\_id}. Check successful responses as well as error handling.

### AI Output Testing

Verify that Gemini produces structured data containing the required story fields. Test handling of invalid JSON, API errors, rate limits, and network failures. Confirm that characters remain consistent and panel continuity is logical according to the project requirement.

### Image Testing

Verify that each requested panel receives an image and that image files are saved under generated/images/. Check that image prompts contain the required character, action, environment, lighting, camera, mood, and art-style information.

### PDF Testing

Verify that the generated PDF contains the title, summary, panel images, narration, dialogue, and page numbers and that it is stored under generated/pdfs/.

### Frontend Testing

Check responsive layout, loading status, comic preview, panel cards, regeneration, PDF download, and New Comic behavior. Verify that Fetch API requests work without unnecessary page reloads.

## Security Testing

Confirm that API keys are not exposed, stack traces are not returned to users, arbitrary local files cannot be accessed, and safe filenames and file paths are used.

*Project reference: ComicCraft – AI Comic Story Creator*